Stellar Object Classification: Separating Quasars, Stars, and Galaxies

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***Abstract*—This project applies machine learning techniques to classify stellar objects from the SDSS17 Stellar Classification Dataset into galaxies, stars, and quasars. Three supervised machine learning algorithms were evaluated: Random Forest, XGBoost, and LightGBM. The dataset was preprocessed by removing non-predictive identifier columns, treating outliers, eliminating highly correlated features, scaling numerical attributes, and applying stratified train-test splitting. The results show that all models achieved high classification accuracy, with optimized LightGBM producing the best overall result. However, quasars remained the most difficult class to classify due to their similarity with galaxies.**

I. Introduction

The fast moving advancement astronomical surveys such as the Sloan Digital Sky Survey has created an era of big data in astrophysics. Telescopes all around the world capture large amounts of photometric and spectroscopic measurements in order to map and locate stellar objects. Classifying those objects into 3 categories such as stars, galaxies, and quasars is an important task for higher understanding of the composition and creation of the universe. However, the large amount of volume and manual classification of the data created from those measurements makes it impossible for accurate and correct classification. This drives scientist into moving towards machine learning approaches.

Our project investigates the application of supervised machine learning techniques used in order to automate the classification of stellar objects using the SDSS17 dataset. By developing an automated and highly accurate multi-model classifier, this project aligns with the United Nation's Sustainability Goals such as SDG9 (Industry, Innovation, and Infrastructure) by creating technological innovation in processing large datasets and SDG17 (Partnerships for the Goals).

In order to address the challenges created by this dataset including the high dimensionality from the features and class imbalance, this study evaluates the performance of 3 powerful supervised machine learning models: Random Forest, XGBoost and LightGBM. Furthermore, the study explores a problem re-framing approach by dividing the multiclass task, and compares our methodology against existing public implementations to validate the robustness of the final model.

II. Literature Review

Recent advancements in computational astrophysics have started to rely heavily on machine learning to process high-dimensional sky survey data. This creates an important task of classifying different stellar objects into different categories such as stars, galaxies, and quasars in order to understand the composition of the universe.

According to Silva et al., who composed a highly relevant study, explores the application of supervised machine learning techniques such as Random Forest and Gradient boosting algorithms for stellar object classification using the Sloan Digital Sky Survey (SDSS) photometric and spectral data [1]. In this study, the authors evaluated an algorithms' ability to classify stellar objects into stars, galaxies and quasars based on specific features such as redshift and ultraviolet-green color indices [2].

Finally, authors concluded that while Random Forest provided a robust baseline with strong resistance to overfitting, boosting models and frameworks offered superior efficiency compared to the high dimensionality of the dataset [3].

Our project expands upon and enhances previous study's methods. Our experimental strategy first involves rigorous hyperparameter tweaking, such as optimizing learning rates, maximum depth, and estimators for LightGBM and XGBoost, instead of depending only on baseline models such as Random Forest. This is followed by a thorough evaluation of the confusion matrices that are created. Second, our work presents a novel technique to problem re-framing that has not been examined in the referenced literature. Our work and research tries to isolate specific classification boundaries by decomposing the typical multiclass classification into sequential binary subproblems, such as Quasar vs. non-Quasar, followed by Star vs. Galaxy, which helps provide a more detailed analytical perspective.

III. New Learning and Data Report

The Stellar Classification dataset obtained from the Sloan Digital Sky Survey (SDSS17) consists of 100,000 observations over 18 features. The three classes in this dataset include Galaxy, Star, and Quasar (QSO). Upon analysis of the data, it was established that there were no missing values within any of the features and no duplicate observations. The class imbalance is moderately skewed, where 59.4% of the data was classified as Galaxies, while Stars and Quasars made up 21.6% and 19.0%, respectively. Each feature in the dataset is numeric. Label encoding was applied to classify the target column, where GALAXY = 0, QSO = 1, and STAR = 2.

Nine identifier columns that have no predictive power since they are administrative identifiers and not physical features were eliminated: obj_ID, run_ID, rerun_ID, cam_col, field_ID, spec_obj_ID, plate, MJD, and fiber_ID. Thus, the dataset was reduced from 17 initial features to 8 useful predictive features.

A. *Preprocessing*

Outlier Treatment: Capping was used to address outliers in all columns by limiting values below the 1st percentile and above the 99th percentile. Unlike removal-based outlier handling techniques, capping preserves all observations while reducing the influence of extreme values during model training.

Correlation Matrix: To remove redundancy among the selected features, correlation analysis was performed. Features with a Pearson correlation greater than 0.95 were eliminated. This left six features for model building.

Feature Scaling: Feature scaling was applied to ensure that each feature contributed equally to the model. All columns were standardized so that their mean became 0 and their variance became 1.

Train/Test Split: Stratified sampling was used to split the data into 80% training data and 20% validation data. This resulted in 80,000 rows for training and 20,000 rows for testing.

Class Imbalance: Considering the large dataset size and the resistance of tree-based methods to slight imbalance, `class_weight="balanced"` was used for classifiers that support it instead of SMOTE, which would be computationally expensive.

IV. Experimental Design and Model Selection

This part of the paper provides details about the goals and methodology used in the experiment phase. The objective of this experiment was to find out whether machine learning algorithms can provide efficient stellar classification using spectral and astronomical data. It means that the goal of this work was the development of a good multiclass model which could distinguish between stars, galaxies, and quasars. To make a reliable classification, it is crucial to minimize classification errors to ensure the maximal quality of predictions.

Firstly, it should be noted that stellar classification plays an important role in modern astronomy and sky surveys. Observations generate huge volumes of data in the form of spectrograms. Due to the vast amount of received data, it is impossible to use manual classification. It means that machine learning is needed to make the process of processing and evaluating data possible and efficient. On the other hand, stellar classification is important in terms of research and exploration because of the possibility to learn something new about the universe. That is why both prediction accuracy and practical usability should be taken into consideration in model creation.

There were several steps that should be completed before starting experimenting with the selected machine learning algorithms. Firstly, the dataset was divided into training and testing sets using the 80/20 split ratio. The experiments showed that 80% of the entire dataset could be used as training set, while 20% were used for testing. It was decided to choose 80/20 ratio for the reasons of convenience and efficiency. Such a ratio allowed creating large training and unbiased test sets.

The selected machine learning models were Random Forest, XGBoost, and LightGBM. The reason for such a choice lies in high efficiency and performance shown by these algorithms when working with high dimensional datasets. First of all, Random Forest is an ensemble method used for combining different decision trees into one model. The algorithm operates by making decisions independently, and then the final result is formed based on all these decisions. The XGBoost algorithm relies on ensemble methods as well and works on the basis of the gradient descent approach. Finally, LightGBM represents another variant of boosting which differs from others because of fast calculations and low memory usage.

K-Nearest Neighbors algorithm was considered during the model selection phase as a possible substitute for the aforementioned algorithms. Nevertheless, it was decided to select Random Forest, XGBoost, and LightGBM because of their efficiency, performance, and ability to deal with complex data. It should be mentioned that ensemble models are much more robust when working with high dimensional data. In addition, ensemble techniques can work more efficiently with unbalanced datasets and data with complex relationships.

Three different types of models were created during the experiments:

- Original model training without preprocessing and hyperparameter tuning;
- Preprocessing and other manipulations prior to the training;
- Optimization of the selected model through hyperparameter tuning.

Thus, it becomes possible to observe differences between models trained on original data, processed data, and data with selected hyperparameters. It should be noted that optimization was needed after the experiment with the preprocessed model to improve performance.

There were several hyperparameters that should be tuned to increase efficiency and reduce the risk of overfitting. These hyperparameters include learning rate, the number of estimators, maximum depth, and number of leaves. The purpose of tuning these hyperparameters is to achieve maximum efficiency and minimize the number of errors.

It is also important to pay attention to different aspects related to performance measurement. First of all, it should be noted that accuracy alone cannot provide relevant information about model performance. That is why it was decided to use several criteria for evaluation, namely:

- Accuracy; measures the proportion of right predictions in comparison with all predictions made by the algorithm.

$$Accuracy = (TP + TN)/(TP + TN + FP + FN)$$

- Precision; measures the proportion of true positive predictions in relation to all predictions.

$$Precision = TP/(TP + FP)$$

- Recall; measures the proportion of identified right predictions in comparison with all positive examples.

$$Recall = TP/(TP + FN)$$

- F1-score; harmonic mean of precision and recall.

$$F1 = 2 \times (Precision \times Recall)/(Precision + Recall)$$

- Confusion matrices

Analysis of the confusion matrix is very important because of mistakes that can negatively influence the accuracy of astronomical observations. The information about the classification performance is not the only thing that can be revealed with its help.

All selected classifiers were evaluated under identical conditions to provide consistent results. Thus, the same train/test split ratio and criteria were used. As a result, it becomes possible to analyze the performance of different models based on confusion matrices.

A. *Results and Confusion Matrix Interpretation*

TABLE I
RANDOM FOREST PERFORMANCE SUMMARY

Configuration	Accuracy	GALAXY F1	QSO F1	STAR F1
Run 1 – No Tuning	97.69%	0.98	0.94	1.00
Run 2 – With Tuning	97.65%	0.98	0.94	1.00

1) Run 1 – No Tuning: The unaltered Random Forest model with `class_weight='balanced'` and `random_state=42` achieved an accuracy of 97.69% on the 20,000 samples in the test set. Across all three categories, the model showed strong predictive performance, with a weighted average precision of 0.98, recall of 0.98, and F1-score of 0.98.

For the STAR category, the model achieved perfect precision and recall scores of 1.00, resulting in a perfect F1-score of 1.00. This indicates that stars were highly separable from the other classes due to their distinct redshift characteristics.

For the GALAXY category, the model achieved a precision of 0.97, recall of 0.99, and F1-score of 0.98. The QSO category was the most difficult to classify, with a precision of 0.96, recall of 0.92, and F1-score of 0.94.

Three key observations emerged from the confusion matrix. First, the most common misclassification involved QSO objects being classified as GALAXY, with 304 quasars misclassified as galaxies. This pattern aligns with scientific understanding because quasars at certain redshifts can share

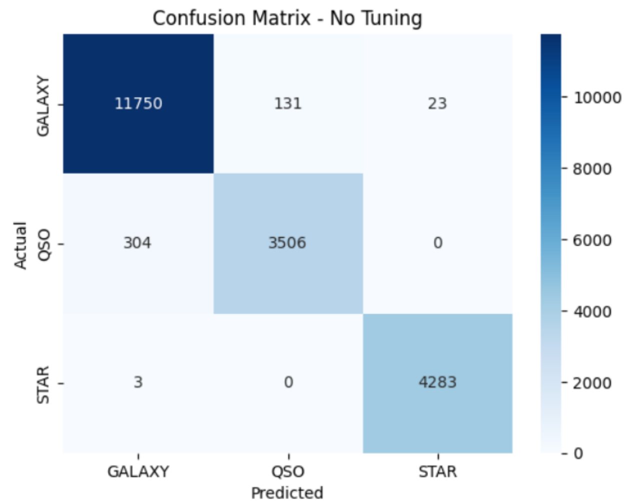

Fig. 2. Random Forest Run 1 Confusion Matrix

brightness-related characteristics with galaxies, making them harder to distinguish. Second, no QSO was misclassified as a STAR, and no STAR was misclassified as a QSO, confirming that STAR objects are spectrally distinct from extragalactic objects. Finally, only three stars were misclassified as galaxies, yielding a star classification accuracy of 99.93%.

2) Run 2 – With Tuning: GridSearchCV was performed using the parameter grid: `n_estimators=`, `max_depth=[None,20]`, and `max_features=['sqrt']` with 3-fold cross validation. The best combination was `n_estimators=300`, `max_depth=None`, and `max_features='sqrt'`, with 97.65% cross-validation accuracy.

The optimized model achieved a test accuracy of 97.65%, which was slightly lower than the default model by 0.04%. The class-level F1 scores remained unchanged, with 0.98 for GALAXY, 0.94 for QSO, and 1.00 for STAR.

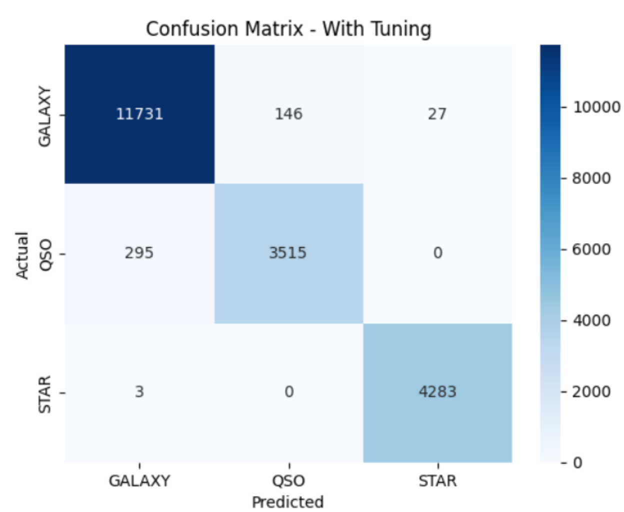

Fig. 4. Random Forest Run 2 Confusion Matrix

Compared with Run 1, the tuned classifier correctly classified 3,515 QSO objects compared to 3,506 in Run 1, reducing QSO-to-GALAXY errors from 304 to 295. However, galaxy classification became slightly worse, with 11,731 correct predictions compared to 11,750 in Run 1. There was also a small increase in GALAXY objects being misclassified as QSO and STAR.

TABLE II
XGBOOST PERFORMANCE SUMMARY

Configuration	Accuracy	GALAXY F1	QSO F1	STAR F1
Run 1 – Baseline	97.52%	0.98	0.94	0.99
Run 2 – Tuned	97.56%	0.98	0.94	0.99

3) Run 1 – Baseline XGBoost: The baseline XGBOOST model with no fine-tuning or hyperparameter tuning achieved a high accuracy of 97.52% on the test data showing that the model demonstrated strong performance across the different object categories with a weighted average precision of 0.98, recall of 0.98 and an F1-score of 0.98.

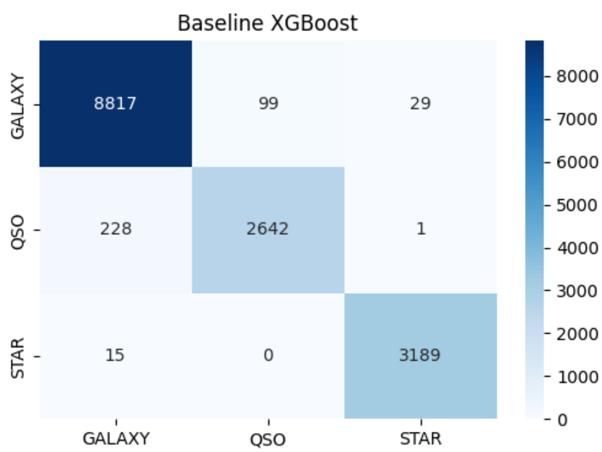

Fig. 5. Baseline XGBoost Confusion Matrix

As revealed in the baseline confusion matrix, it shows that the XGBOOST model correctly classified 8,817 GALAXIES, 2,642 QSO, and 3.189 STARS. The class/object that was frequently misclassified happened with the QSO objects being predicted as GALAXY.

4) Run 2 – Tuned XGBoost: Although the baseline model achieved an overall accuracy of 97.52% on the test data, we still needed to apply hyperparameter tuning to the XGBOOST model in order to check if an even higher accuracy can be achieved because AI models are required to be as accurate as possible. Hyperparameter tuning was conducted using the import RandomizedSearchCV with a 3-fold cross-validation. The search grid included variations in `n_estimators`, `max_depth`, and `learning_rate`. The optimal parameters identified were `n_estimators=200`, `max_depth=9`, and `learning_rate=0.2`, which yielded a best cross-validation score of 97.53%.

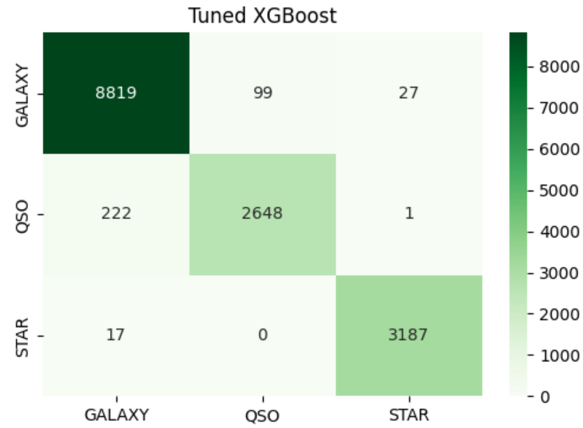

Fig. 6. Tuned XGBoost Confusion Matrix

The tuned model's confusion matrix demonstrates minor improvements in distinguishing the challenging classes. The number of correctly classified QSO objects increased to 2,648, while the number of QSO objects misclassified as GALAXY decreased from 228 to 222. The model correctly predicted 8,819 GALAXY objects and 3,187 STAR objects. As observed across the other models, no QSO objects were misclassified as STARs, confirming the distinct spectral separation of stars from extragalactic entities.

TABLE III
LIGHTGBM PERFORMANCE SUMMARY

Configuration	Accuracy	GALAXY F1	QSO F1	STAR F1
Run 1 – No Preprocessing	97.63%	0.98	0.94	1.00
Run 2 – With Preprocessing	97.63%	0.98	0.94	1.00
Run 3 – Tuned LightGBM	97.775%	0.98	0.94	1.00

5) Run 1 – No Preprocessing: A LightGBM model was used without any preprocessing and directly trained on the original dataset. It was designed as a multiclass classifier with a three-class output and random state 42. LightGBM reached an accuracy of 97.63% when tested on the dataset with 20,000 records. In terms of metrics, the LightGBM model showed great performance on all categories with weighted average precision of 0.98, recall of 0.98, and F1-score of 0.98.

In the STAR category, the classifier showed perfect values for precision and recall equal to 1.00, which resulted in a perfect F1-score of 1.00. It indicates that stars have specific features that help distinguish them from other objects such as galaxies and quasars. When classifying objects of the GALAXY category, LightGBM showed a precision of 0.97, recall of 0.99, and F1-score of 0.98. Similarly to the Random Forest model, the QSO category proved to be the most difficult to classify, having a precision of 0.96, recall of 0.92, and an F1-score of 0.94.

There are three key points that can be seen from the confusion matrix. Firstly, the most frequent mistake made by LightGBM was that the QSO objects were classified as GALAXY objects. There were 294 quasars classified as

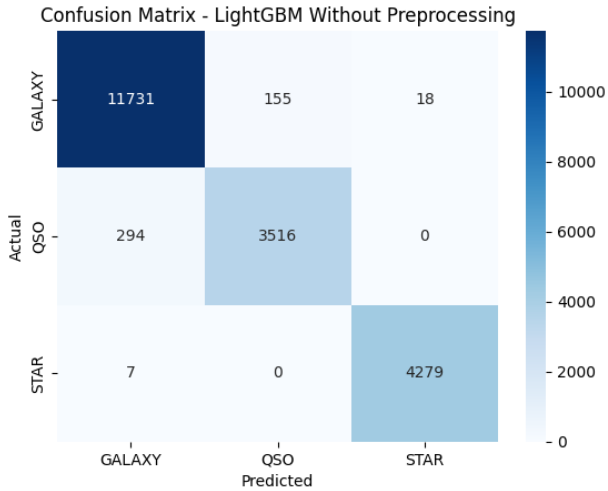

Fig. 7. LightGBM Run 1 Confusion Matrix

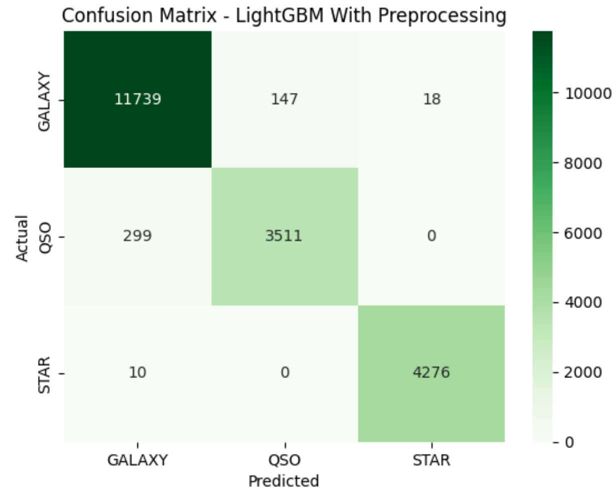

Fig. 8. LightGBM Run 2 Confusion Matrix

galaxies in the testing dataset. As was previously mentioned, such misclassifications are consistent with astronomical observations as quasars may look like galaxies due to some redshift value. Secondly, there were no cases of QSO objects being classified as STARS or vice versa. It proves once again that it is easier to separate stars from extragalactic objects. Thirdly, only seven stars were classified as galaxies, thus proving that stars can be recognized with extremely high accuracy. Overall, the confusion matrix shows the great ability of the LightGBM classifier to recognize objects of the stellar category while keeping strong performance on two other categories.

6) Run 2 – With Preprocessing: The next experiment with LightGBM was performed with the use of preprocessed data. During the data preprocessing, outliers were treated, a correlation analysis was performed and irrelevant features removed, and features were scaled and train-test split performed with stratification. After the preprocessing steps, LightGBM showed accuracy equal to 97.63%, which is the same result as when the model was trained on the original dataset. In addition, the values of weighted average precision, recall, and F1-score were not affected. Such results indicate natural robustness of LightGBM to different types of data processing.

On a class level, F1-score for GALAXY objects equals 0.98, F1-score for QSO objects equals 0.94, and perfect F1-score of 1.00 was received for the STAR category. Even though the overall accuracy was the same, several minor changes can be seen in the confusion matrix.

The amount of correctly classified GALAXY objects rose to 11,739 from 11,731 in Run 1. At the same time, the amount of QSO objects that were correctly classified decreased slightly and became 3,511 instead of 3,516. The STAR category was very stable, with only 10 stars classified as galaxies. As in the first run, there was no case of STAR being misclassified as QSO and vice versa.

Such results show that the preprocessing had very little effect on performance of LightGBM as this model can successfully work with the original dataset as well. This also highlights the robustness of this type of classifier when dealing with astronomical datasets of considerable size.

7) Run 3 – Tuned LightGBM: RandomizedSearchCV was used with 3-fold cross-validation and several parameters of the LightGBM model were optimized. The randomized search considered possible values for learning rate, estimators, max_depth, and num_leaves. The best combination of parameters found during the search included 300 estimators, learning rate 0.1, max_depth 10, and 63 leaves. The best cross-validation score obtained in this step was 97.31%.

The optimized model achieved a test accuracy of 97.775%, which became the highest result achieved among all classifiers evaluated in this study. The class-level F1 scores remained very strong, with 0.98 for GALAXY, 0.94 for QSO, and 1.00 for STAR.

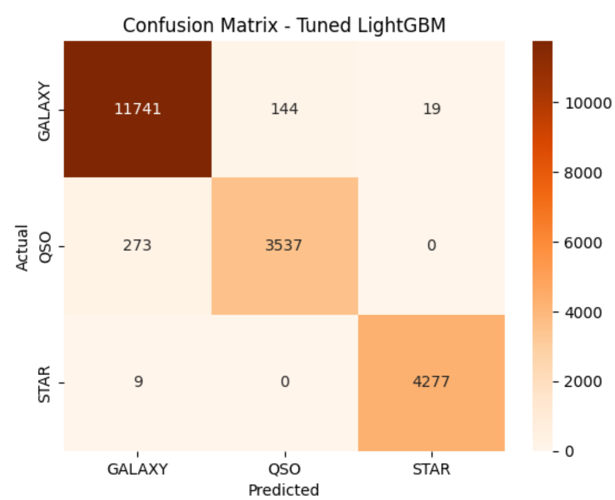

Fig. 9. Tuned LightGBM Confusion Matrix

Compared with the previous runs, the performance of

LightGBM improved slightly after hyperparameter tuning. The tuned model correctly classified 11,741 GALAXY objects and 3,537 QSO objects. Furthermore, the number of QSO objects misclassified as GALAXY objects decreased to 273 compared to 294 in Run 1 and 299 in Run 2. Since quasars represented the most challenging class throughout the project, this improvement is particularly important.

Similar to the previous experiments, no QSO object was misclassified as a STAR and no STAR object was misclassified as a QSO. Only nine STAR objects were incorrectly classified as GALAXY objects, confirming that the STAR category remained highly distinguishable from the other classes. The confusion matrix therefore demonstrates that parameter tuning improved classification performance while maintaining strong predictive capability across all categories.

Overall, these results indicate that the LightGBM model benefits from hyperparameter tuning. The optimized LightGBM classifier achieved the highest accuracy among all evaluated models, reaching 97.775%, which makes it the best-performing classifier in this study.

B. Discussion and Model Comparison

Three selected models (Random Forest, XGBoost, and LightGBM) were evaluated using identical splitting ratios and criteria. Random Forest showed high robustness and performed stellar classification with high accuracy while avoiding overfitting with the help of ensembles. XGBoost also proved its high performance with the help of boosting and optimization process.

Nevertheless, LightGBM showed the highest quality and performance among all models discussed here. In addition to high values of the mentioned metrics, it was the most efficient and fast boosting classifier which was easy to tune. The analysis of confusion matrices also showed that the model made few mistakes during classification of galaxies, stars, and quasars.

As it can be seen, the use of ensemble and boosting techniques is effective when dealing with large, complex, and high dimensional datasets. Preprocessing and hyperparameter tuning significantly improve performance and robustness of models. Thus, it is safe to say that LightGBM is the best performing model in this project.

C. Problem Re-Framing: Deep Space Isolation (Galaxy vs. Quasar)

Because stars are easily detected due to their unique features, their inclusion in the training artificially inflates multiclass accuracy and hides the true accuracy of the dataset. To deal with this challenge, we removed all STAR observations, re-framing the task as a binary classification problem focused on the differentiation of the Galaxy VS. Quasar classes.

Training the tuned LightGBM model on this modified dataset showed an accuracy of 97.00%, with F1-scores of 0.98 for GALAXY and 0.94 for QSO. Without the star data inflating performance, the model focused on the binary class results instead.

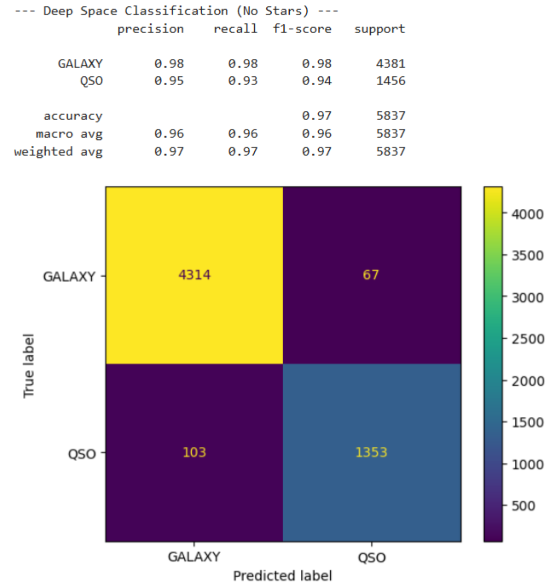

Fig. 10. Galaxy Vs. Quasar Model Classification

D. Kaggle Notebook Comparison

For this part of the project, a public Kaggle notebook was selected and studied (https://www.kaggle.com/code/navdeepjakhar/stellar-classification-comparing-6-algorithms). It also deals with stellar classification. However, there are significant differences between it and the implemented project.

Firstly, our implementation focuses on multiclass classification and uses more models. There are several ensemble models like Random Forest, XGBoost, and LightGBM. In the selected notebook, different models were compared as well. However, our work is more focused on comparison and preprocessing of data.

Secondly, the Kaggle notebook contains a limited set of experiments. Our project includes many stages and models with the purpose of analyzing and comparing classifiers. There is no preprocessing and hyperparameter tuning in the selected notebook. Therefore, it provides a very basic approach towards solving this task.

Thirdly, the interpretation of experimental results is completely different in these two projects. Confusion matrices were used in both cases to evaluate the performance of the models. However, in our project, the matrices were analyzed in detail to identify mistakes and problems related to classification.

V. Impact

This project has several important academic and practical impacts. From a scientific perspective, automated stellar classification supports astronomers in processing large-scale astronomical survey data more efficiently. Since modern surveys generate massive amounts of data, machine learning can reduce the need for manual classification and help researchers identify galaxies, stars, and quasars more accurately.

From a technological perspective, the project demonstrates how machine learning can be used to solve real-world classification problems involving large and complex datasets. The use of ensemble models such as Random Forest, XGBoost, and LightGBM shows how advanced algorithms can improve decision-making and pattern recognition in scientific datasets.

The project also supports innovation in data-driven research. By comparing several models, applying preprocessing, and analyzing confusion matrices, the study provides a clearer understanding of where machine learning performs well and where it still faces limitations. In particular, the confusion between quasars and galaxies shows that some classification errors may come from the nature of the data itself rather than from the weakness of the model.

VI. Conclusion

In this project, three machine learning algorithms, were applied to the SDSS17 Stellar Classification Dataset , Random Forest, XGBoost, and LightGBM, to classify galaxies, stars, and quasars using 100,000 stellar observations. The preprocessing pipeline included removing non-contributive identification features, outlier treatment, correlation-based feature elimination, standardization, and stratified training-validation splitting.

The accuracy for all three classifiers was impressively high, ranging from 97.53% to 97.775% across six trials. Stars proved to be the easiest class to classify, achieving almost perfect classification performance in all trials. The greatest classification difficulty was observed with quasars due to their similarity to galaxies, which has also been noted in astronomical observations. This ambiguity arises because extragalactic objects cannot always be clearly identified based only on luminosity-related features.

Among the classifiers used, the optimized LightGBM model achieved the highest accuracy of 97.775%. LightGBM also showed robustness to the preprocessing process, producing the same results before and after applying it. Tuning Random Forest and XGBoost did not improve accuracy and instead slightly reduced performance. This suggests that the default settings of these classifiers already provided strong generalization performance on this dataset.

The QSO-GALAXY confusion observed across all models suggests a fundamental data limitation rather than a modeling issue. Future work could explore deep learning techniques using raw spectra, improved feature extraction methods designed to capture quasar characteristics, or reframing the task as a binary classification problem to predict whether an object is a quasar or not. Another possible experiment would be to remove the redshift feature completely and evaluate how strongly the models depend on it.

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